

Research Statement

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Asset prices tell firms where to invest and help determine how much they can borrow. I study how these two roles shape capital allocation and welfare, and how artificial intelligence and data change them. My work in macro-finance asks when asset values and financial expertise expand firms' borrowing capacity, and whether the resulting investment improves allocation and welfare. My research on AI and data examines how shared information shapes economic decisions and how automation changes wages, saving, and welfare across generations. My job market paper connects these interests through the informational role of asset prices: a shared AI signal can improve investors' forecasts while making prices a less informative guide for firms' investment. I use general equilibrium theory, empirical evidence, and quantitative analysis.

Job market paper

My job market paper, *Common AI Information and Capital Allocation*, asks how a shared AI information source changes financial markets' ability to allocate capital. Private research errors can average out through trading, while AI model errors are shared across investors researching the same firm and can survive aggregation into prices.

I use a large language model to analyze earnings calls to motivate this mechanism. Manager pairs whose trades show greater common alignment with this information subsequently hold more similar portfolios, controlling for existing similarity. The relationship is stronger after ChatGPT's public release and where the model has a clearer information advantage.

I embed this mechanism in a macro-finance model in which investors choose how much information to buy and how to divide it between private research and the AI signal. Their trading sets prices. Firms read those prices and choose capital. Private errors are specific to each investor, so prices average them out. The AI error is common, so it stays in the price. Investors may underestimate the dependence between their AI signal and information already reflected in prices. In equilibrium, capital could move toward the firms the model favors, not necessarily toward the more productive firms. Aggregate productivity changes because capital is reallocated.

AI advances can improve investors' forecasts while making prices less informative. Cheaper access and greater accuracy have different effects. When AI is so expensive that its use is negligible, a small cost reduction raises aggregate productivity: the information gain outweighs the increase in exposure to common errors. By contrast, when use is negligible because the model is inaccurate, an accuracy improvement can lower productivity, since greater reliance can amplify common error exposure more than additional informed trading reduces price noise. Recognizing common errors mitigates this force. Numerical equilibria illustrate a J-shaped relationship between accuracy and productivity and a hump-shaped response to cheaper access.

The paper connects the real effects of financial markets with work on information and misallocation, including David, Hopenhayn, and Venkateswaran (2016). Its contribution is to make investors' exposure to a common error endogenous. Individual forecast precision alone is then insufficient to assess whether better information technology improves aggregate allocation.

Asset values, financial frictions and allocation

My macro-finance papers examine the financing role of asset values: how much future income firms can pledge, how that capacity develops, and whether additional borrowing improves allocation and welfare.

In *Bubbly Booms and Welfare*, with Feng Dong and Yang Jiao (*Review of Economic Dynamics*, 2024), we show that higher housing values relax firms' borrowing constraints, but housing construction draws labor from other sectors. With sufficiently strong production externalities, the reallocation lowers welfare despite the financing benefits. *Contagious Bubbles*, with Feng Dong, Zhen Huo, and Yi Wen (minor R&R, *Journal of Monetary Economics*), studies how financial links allow one sector's bubble to support valuations elsewhere. The resulting contagion is asymmetric: a critical sector's collapse can bring down others without the reverse occurring. Our quantitative analysis identifies the financial sector as a potential source of this fragility.

Financial development also changes collateral values. In *Financial Development and Housing Kuznets Curve*, with Zhifeng Cai and Feng Dong (R&R, *International Economic Review*), we document a hump-shaped relationship between financial development and house price-to-rent ratios. A model with occasionally binding borrowing constraints and capital and land as collateral explains this pattern: during the transition from a constrained to an unconstrained steady state, the ratio can exceed its level in either steady state.

The capacity to pledge future revenues also depends on financial expertise. In *Schumpeterian Innovation and Financing Cycles*, with Yi Dong and Zhenyang Xu, expertise accumulates through financing activity and supports further innovation. This feedback can sustain multiple growth paths and give temporary research disruptions persistent effects. Sectoral differences in financing capacity matter for policy as well. In *A Macroeconomic Model of Structural Monetary Policy in China*, with Feng Dong and Dongzhou Mei (*International Economic Review*, 2025), we show that credit tightening disproportionately raises private firms' financing costs; structural monetary policy can address the resulting misallocation.

AI, data, and welfare

My research on AI and the data economy examines both labor substitution and information production. The first changes wages, saving, and welfare across generations; the second changes the information available for economic decisions.

In *Automation and Dynamic Efficiency: When Do Robots Benefit Every Generation?*, with Feng Dong, Ernest Liu, and Pengfei Wang, we model robots as both productive inputs and assets through which young households save. Wage-compressing automation raises asset returns while reducing the income from which households accumulate wealth. The transition generation gains on existing assets, whereas later generations face lower wages and higher returns. At the onset of adoption, a marginal improvement in robot technology benefits every generation if and only if the economy without robots is dynamically inefficient. Once robots are in use, the criterion compares the saving base that receives return gains with the present value of wage losses. If the initial economy is dynamically efficient, the equilibrium with robots remains Pareto optimal. In this setting, automation policy redistributes welfare across generations and cannot make every generation better off.

My published work with Jinpu Li and Feng Dong (*China Economic Quarterly*, 2026) studies how these tensions differ across households and cohorts. In an overlapping-generations model with saving households and hand-to-mouth households, AI substitution across tasks widens income gaps and imposes human-capital losses on new cohorts. Restricting the tasks AI may perform limits those losses and also limits the output gain from the technology. The paper analyzes the redistributive effect and policy trade-offs in such an economy.

On the information side, *Big Push Towards Big Data*, with Feng Dong and Jinpu Li, studies how production generates data that improve demand forecasts and support further production. This feedback can trap the economy at low data and low output. Temporary public information provision can help it escape, although producing signals uses labor and can crowd out private activity when information is abundant. Our related paper, *The Data Trap: When AI Fails*, examines how reliance on synthetic data can undermine data quality and subsequent AI development.

Ongoing work and research agenda

My next projects bring together financing constraints and the macroeconomics of AI. In ongoing work with Feng Dong and Neng Wang, *AI Valuations and Automation*, financially constrained AI producers receive cash flows from sectoral knowledge spillovers, independently of their own installed capital. Under limited commitment, these cash flows support a fully pledgeable component of firm value beyond the value of installed capital. Better pledgeability of installed capital raises investment but lowers value per unit of capital, so firm value relative to human capital can fall as financing improves. Allowing AI to take over additional tasks expands demand for its services and amplifies the investment response to easier financing. Task reallocation also strengthens the valuation response to higher AI productivity; with fixed tasks and strong complementarity, productivity gains can instead reduce the AI sector's size and value relative to human capital. Next I want to study reversals in expected technology, and to compare financing arrangements.

Building on my job market paper, I plan to examine how price informativeness and firms' investment responses to prices change as language models become more accessible. I plan to use the price-informativeness measures in Bai, Philippon, and Savov (2016) and the sensitivity of investment to the stock price in Chen, Goldstein, and Jiang (2007), and to compare firms by call length, analyst coverage, and how decisive a model score of the call is. This would help assess how the effects vary across information environments, without identifying individual managers' adoption decisions. The theory's distinction between cheaper access and greater accuracy would guide the interpretation of these patterns.

In theory, I plan to combine the two strands of my research and let one price do both jobs. A price that contains a common AI error changes what a firm infers about its fundamentals and how much it can pledge. A noisier price both sends capital to the wrong firms and may tighten borrowing. I want to know when a financing constraint makes the misallocation worse and when it offsets the misallocation.